

Enterprise Project

Year 12 Higher School Certificate Enterprise Computing Task 3



HURLSTONE
AGRICULTURAL HIGH SCHOOL

Name: William Lam

Class: 12ECOMA

Teacher: Ms Hidalgo

Due Date: 30/06/2026

Table of Contents

Problem Definition and System Proposal.....	3
Enterprise Context.....	3
Problem Definition.....	3
Proposed System.....	4
Project Plan.....	4
System Design and Modelling (5 Marks).....	4
System Modelling.....	4
Data Flow Diagram (DFD).....	4
System Flowchart.....	5
Data Modelling.....	6
Schema.....	6
Entities.....	6
Table: Users.....	6
Table: Sessions.....	6
Table: Parts.....	7
BRIDGE Table Interactions.....	7
BRIDGE Table: Orders.....	7
Entity Relationships.....	7
ERD.....	8
Data Dictionary.....	8
Users.....	8
Sessions.....	9
Parts.....	9
Interactions.....	10
Orders.....	10
Testing, Evaluation and Impact.....	11
Testing.....	11
Test Scenarios.....	11
Checking Against Original Requirements.....	12
Example User Interaction.....	12
Evaluation.....	13
Strengths.....	13
Limitations.....	13
Improvements.....	13
Social, Ethical and Legal Impacts.....	13
Data privacy.....	13
Security of information.....	14
Bias in automated systems.....	14
Impact on staff or users.....	14
Future Improvements.....	15
Appendix A – Client Interview.....	15

Problem Definition and System Proposal

Enterprise Context

The organisation is an automotive dealer which includes a workshop that specialises in maintaining its vehicles. Mechanics serve as the central figures in day to day operations, operating with customers and their vehicles to ensure that their requirements are met. Vehicles each have their own parts and different methods of maintenance, meaning mechanics need to be well versed in operating on a wide variety of vehicles. In the backend, stock takers ensure that inventory is kept to a high standard, and no parts are missing to avoid delays in servicing vehicles. Each month, hundreds of different vehicles come in for service, meaning hundreds of parts need to be used and sometimes ordered. That is the stock taker's job. Mechanics will communicate with stock takers to ask for new parts, and stock takers will either give them access to the part if it is on hand, or go to suppliers for delivery if necessary. This chain allows for effective servicing of cars, ensuring that customers are satisfied with their service.

Problem Definition

In the digital age, this organisation wants to modernise its methods of ordering parts for its vehicles. This is to ensure it keeps up with its competitors, ensuring it stays at the top. Going digital allows for faster and more efficient communication between the three parties (mechanic, stock taker, supplier) which will have the carry on effect to its customers who will experience faster service.

The client, who is a mechanic, wants a mobile application that can identify vehicle parts using a camera whilst under a car, and provide audio feedback about what it is. It must also have an interface within that app to request parts to the stock takers, reducing friction between the two. It is to be aimed at experienced mechanics, rather than new employees. This is to ensure that new employees are taught to not have a reliance on the application for identification of vehicle parts.

The system will be designed for use by mechanics only, as a system for actually ordering parts from the stock takers already exists. It must merely be able to send requests for new parts, not order them.

Functional	Non-Functional
Can connect to external devices Uses camera to identify vehicle parts Provides audio/visual feedback Can request parts directly within the app	Clean, minimalist UI Smooth Animations Uniform Colour Scheme

Proposed System

Link to Prototype:

<https://www.figma.com/proto/IQEhEVizCWXGp7NdPigue8/Assessment-Task-3?node-id=45-394&p=f&t=itylFLPknnR3hU2B-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=45%3A394>

The proposed system will be a mobile application that will be designed using Figma. Its name will be "What Am I Looking At?", or WAILA for short. Figma is a cloud based design tool that can be used to work on websites and/or applications. Designs can be shared, meaning it holds a unique position in allowing the client and developer to work together to achieve the required goals.

The application will use a camera to look at vehicle parts, where an AI will then identify it, and bring up an overlay describing the part as well as use speakers to verbally convey the information. It can be connected to wireless devices such as headphones/speakers, and external cameras such as gopros.

Identified vehicle parts will be highlighted on displays, and if clicked, the client can view its description as well as whether or not it is in stock. From there the client can request for it, with an automatic request sent to stock takers if it is not in stock.

Project Plan

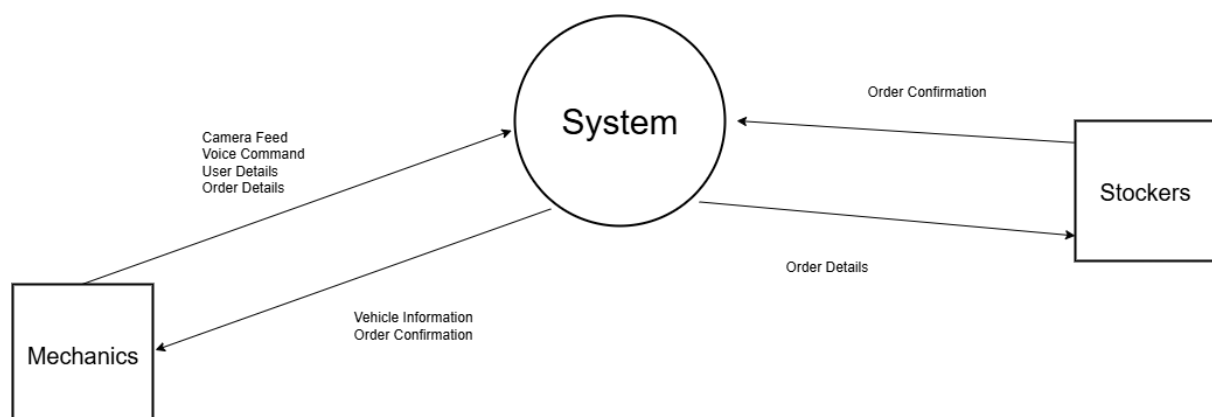
Gantt Chart has been uploaded separately as an Excel spreadsheet. Milestones are the stages that are *italised*, and **bolded**.

System Design and Modelling (5 Marks)

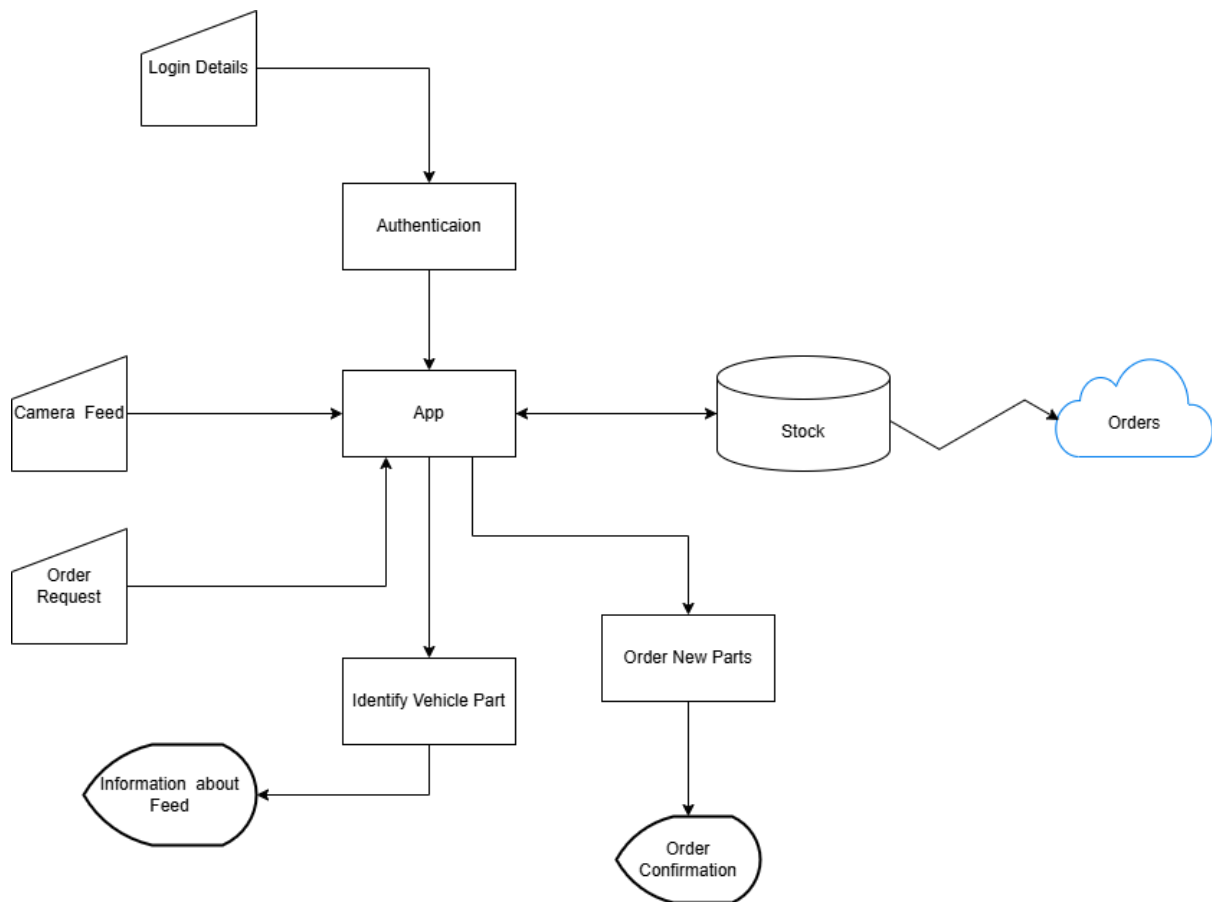
System Modelling

Data Flow Diagram (DFD)

This is a Level 0 DFD.



System Flowchart



Data Modelling

Schema

Entities

- Users
- Sessions
- Parts
- Interactions
- Orders

Table: Users

PK: User_ID

Fields:

- Name
- Phone
- Date_Signed_Up

Table: Sessions

PK: Session_ID

Fields:

- Date
- Title
- Photo
- Description

Table: Parts

PK: Part_ID

Fields:

- Name
- Vehicle
- Description

BRIDGE Table Interactions

PK: Interaction_ID

FK: User_ID → Users

FK: Session_ID → Sessions

Fields

- Date
- Time

BRIDGE Table: Orders

PK: Order_ID

FK: User_ID → Users

FK: Part_ID → Parts

Fields:

- Date

Entity Relationships

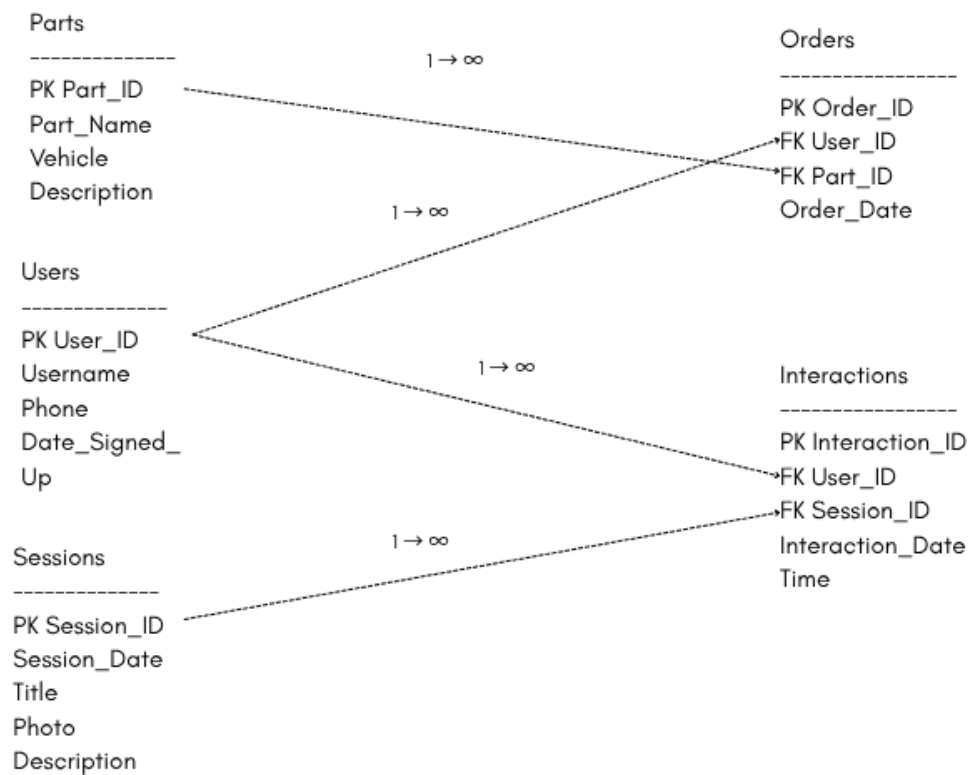
Orders connect Parts and Users, hence it is:

- Users **many** → **many** Parts
 - *Many Users can have many Parts through Orders*

Interactions connect Users and Sessions, hence it is:

- Users **many** → **many** Sessions
 - *Many Users can have many Sessions through Interactions*

ERD



Data Dictionary

Note: An access database has been created which mimics the data dictionary below.

Users

Field Name	Data Type	Data Format	Field Size	Description	Example
User_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the user	1
Username	Short Text	XXXXXX	20	Name of the user	William Lam
Phone	Short Text	XXXXXX	10	Phone number of the user	0412345678
Date_Signed_Up	Date/Time	YYYY/MM/DD	8	Date of when the user	2025/06/07 1605

				registered.	
--	--	--	--	-------------	--

Sessions

Field Name	Data Type	Data Format	Field Size	Description	Example
Session_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the session	1
Session_Date	Date	YYYY/MM/DD	8	Date of the session	2026/06/14
Title	Short Text	XXXXXX	15	Title that describes the session	Tire
Photo	BLOB (Binary Large Object)	PNG/JPEG	5 mb	Photo of the vehicle part	 (Taken from link)
Description	Short Text	XXXXXX	50	More detailed explanation of the session	Allows the vehicle to move.

Parts

Field Name	Data Type	Data Format	Field Size	Description	Example
Part_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the part	1
Part_Name	Short Text	XXXXXX	15	Name of the vehicle part	Tire
Vehicle	Short Text	XXXXXX	15	Vehicle the part is from	Tesla Cybertruck
Description	Short Text	XXXXXX	50	Explanation of what the	Allows the vehicle to

				part is and does	move.
--	--	--	--	------------------	-------

Interactions

Field Name	Data Type	Data Format	Field Size	Description	Example
Interaction_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the interaction	1
User_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the user	1
Session_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the session	1
Interaction_Date	Date	YYYY/MM/DD	8	Date of when the interaction occurred	2026/03/02
Time	Time	HHmm	4	Time of when the interaction occurred in 12 hour time	0801

Orders

Field Name	Data Type	Data Format	Field Size	Description	Example
Order_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the order	1

User_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the user	1
Part_ID	Autonumber	NN	-2,147,483,648 to +2,147,483,647	Computer generated number that identifies the part	1
Order_Date	Date	YYYY/MM/DD	8	Date of the order	2025/04/23

Testing, Evaluation and Impact

Testing

Link to Prototype:

<https://www.figma.com/proto/IQEhEVizCWXGp7NdPigue8/Assessment-Task-3?node-id=45-394&p=f&t=itylFLPknnR3hU2B-1&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=45%3A394>

Test Scenarios

The prototype was tested, first in each module (listed below) and then tested as a whole system.

TEST	RESULT
Login and Registration Screen	PASS
Verification Screen	PASS
Connect External (Camera, Speakers, Internet)	PASS
Camera Screen	PASS
Camera Screen opens text overlays	PASS
Text overlays can request parts	PASS
Parts Confirmation Screen	PASS
Logout	PASS
Whole System	PASS

All buttons, overlays etc were tested thoroughly and edge cases were tested, which led to a few bugs but those were removed.

Checking Against Original Requirements

Original Requirements

Functional	Non-Functional
Can connect to external devices Uses camera to identify vehicle parts Provides audio/visual feedback Can request parts directly within the app	Clean, minimalist UI Smooth Animations Uniform Colour Scheme

Testing

REQUIREMENT	RESULT
Can connect to external devices	PASS
Uses camera to identify vehicle parts	PASS
Provides audio/visual feedback	PASS
Can request parts directly within the app	PASS
Clean, minimalist UI	PASS
Smooth Animations	PASS
Uniform Colour Scheme	PASS

The system has met the set requirements to a satisfactory standard, which will satisfy the client's needs.

Example User Interaction

An example of the entire system process which goes through the mechanic requesting a part, the stock taker asking a supplier, and the supplier delivering the part, is outlined in the cisco packet tracer file, which has been uploaded separately.

Example Process (Mechanic):

Open App → Login → Verify → Connect External Camera → Open Scanner → Take Photo → View text overlays → Request Parts → Confirmation → Home Page → Logout

Evaluation

Strengths

The system has accurately portrayed what the client wishes. It is simple, yet extremely effective. This makes the user experience intuitive. It has a very linear path when using the application (login, press scan, information pops up, request parts etc). This effectively solves the problem the client has. By digitising the process of requesting parts, as well as using machine learning to identify vehicle parts, the organisation will be able to remain ahead of the competition in terms of speed and efficiency.

Limitations

Unfortunately this system has no real proper functionality, instead it is essentially a high fidelity prototype. This means a proper backend cannot exist. This figma design serves merely as a way to show clients what a final product may be, but it is not ready for deployment.

This means that data is not saved between sessions and is hence not a proper functioning system. However, given that a prototype is the only thing required, it serves its purpose well.

Improvements

After going back to the client after a draft prototype had been made, they pointed out clear points of improvement, specifically a change in the login system and the target audience. For the login system, I transitioned from a traditional username/password scheme to a more contemporary verification code system. Now, the users do not require a password, instead just a phone number and the verification code sent via SMS to login. For the target audience, the original was set to all mechanics, new and experienced. However, given the client's intentions for this to be an aid, not a trainer, the audience has been shifted to more experienced mechanics, to provide help, rather than do the job for them.

Social, Ethical and Legal Impacts

Data privacy

WAILA collects and stores personal data about its users, including their phone numbers, session history and part requests. According to the Privacy Act 1988 and the Australian Privacy Principles (APPs), users must be informed specifically about what data is collected, why it is collected, and how it is stored. As phone numbers are stored for SMS verification, this is classified as sensitive personal information, and therefore a privacy policy must be released and agreed to by users before they can use the system. Data should only be retained if absolutely necessary, and deleted if otherwise. Session photos can raise further privacy concerns, as they may capture

other workers, the workshop environment, or proprietary vehicle information. Automated deletion schedules should be in place to remove old data after a set amount of time.

Security of information

WAILA, as part of its system, transports part requests between mechanics and stock takers, handling sensitive business data. This data must be encrypted to avoid external parties from accessing this data. If this data were to be breached, hackers could gain access to the organisation's inventory levels, supplier relationships or vehicle servicing patterns, all of which will be confidential information. All data should be encrypted using HTTPS/TLS protocols. Furthermore, for the SIM-based verification system, rate limiting should be implemented on the number of login attempts, and unusual locations should be flagged for logins. This can lead to an ethical impact where an unauthorised user may gain access to the original user's phone number, allowing them to impersonate them. Legally, the company must be held to the Notifiable Data Breaches (NDB) scheme, meaning that if the system were to be breached, the Office of the Australian Information Commissioner (OAIC) must be notified immediately, as well as its users within 24 hours. Failure to comply will result in legal action as well as financial penalties.

Bias in automated systems

The machine learning in WAILA is only as reliable as the data it was trained on, meaning built-in bias can have real consequences in a professional environment. The AI must be able to reliably identify parts, as the mechanic would then act on that information. If that information is wrong, the organisation could face liability under occupational health and safety legislation, particularly if it leads to dire consequences such as a vehicle being returned in an unsafe condition or further workplace accidents. WAILA is officially a support tool, not an entire replacement, meaning that mechanics should rely on it for a second opinion, not to make the final decision themselves. On the ethical side, if the model is trained predominantly on common passenger vehicles, it will perform poorly on commercial trucks or heavily modified parts, which can create an uneven experience across different mechanics. There should be a way of indicating how sure the AI is. As the client said, a social impact of using WAILA would be the risk of devaluing skilled knowledge. If WAILA were to be given to new or less experienced staff, the organisation will begin to substitute genuine expertise with a reliance on AI, weakening the professional knowledge base of its workforce over time.

Impact on staff or users

By integrating WAILA into the organisation, it will severely change daily workflows for both mechanics and stock takers. As the system requires a personal phone number for SMS verification, this can raise concerns under the Privacy Act 1988. Workers may not want to give their phone number to the app, and that can lead to conflict within the

workplace. To combat this, company issued phone numbers could be used to resolve the idea of using personal numbers. The mechanics and stock takers will need to have time to get used to the new system, which can at first lead to a drop in efficiency, but over time that will resolve as the system is very intuitive and easy to use. On the ethical side, replacing verbal par requests with a logged digital system changes the dynamic between the mechanics and management. As each request is now digitised, time-stamped, and tied to a user's identity, it creates the risk that the system is repurposed as a performance monitoring tool without staff knowledge or agreement. The workers have a right to transparency about how their data is used, and as such, if a change like this were to ever occur, the workers must be informed first. Socially before WAILA. Informal communication between the mechanics and stock takers serves as a way to build trust and contribute to a cohesive team culture. Replacing this with a digital-only interface risks eroding the collaborative environment that effective workshops depend on.

Future Improvements

As this is a prototype, a large number of proposed features could not be properly implemented. These include a proper backend with authentication, input validation and actually training and loading an AI model for the machine learning component of the app. All of these would contribute greatly to the finalised production-ready product for the clients. Cloud syncing would also be another great feature, so mechanics and stock takers do not have to work solely on one device. This would also allow data to be transferred between different users, which will overall lead to a better collaborative environment. Additionally, when viewing past part requests, more information could show up rather than just whether it is in process, denied, or fully approved. For the overlays when the photo is analysed for vehicle parts, a web search function could be implemented just to provide a wider scope of information about that part. A great use of this would be to integrate it with an image searching engine such as Google Lens, which will search for similar photos and provide information about them. Furthermore, actually having visual feedback, which is something that Figma does not support, would be highly beneficial in the final product as this allows users to work entirely hands-free, which can boost safety and productivity.

Appendix A – Client Interview

Note: Below appendix has been recreated, it is not word for word.

What is your organisation and what is your role at said organisation?

I work in the workshop at [REDACTED] dealer, who sells and services vehicles. My specific role is a mechanic apprentice working directly with the vehicles.

I'm looking to create a web-based solution to a problem, however I don't currently have any worthwhile problems worth solving. Does your organisation currently need anything?

My organisation currently wants to modernise itself, as our upper executives feel that we are falling behind the competition.

What do you want your proposed solution to be?

I would like it to be a mobile app, one that I can install on my phone and it needs to allow me request parts directly from the app, instead of me going to a stock taker for a request.

Got it. I'm thinking of using the phone's camera to view a vehicle's underside and then identify vehicle parts. From there you can click on certain elements and see information about them. You can also request parts from that same screen, which will send a request directly to the stock taker.

Great, that sounds good.

After the initial prototype was made, the client was approached for feedback.

Do you have any feedback regarding the prototype?

Regarding the login page, could you make the login use a verification code instead of a password? Just to simplify things.

Easy, I can make that happen. Anything else?

The design is good, very straightforward. I just want to reiterate that the target audience is for senior mechanics, not juniors. I don't want new employees to grow a reliance on using the app instead of their own intuition.

Absolutely, I get that. Thank you for your time.